American Computer Science League

2020-2021 ● Contest 3: Shorts ● Senior Division

1. Boolean Algebra

Identify the ordered triples that make the following expression FALSE?

$A\overline{B} + \overline{BC} + \overline{A}C$

A. (1,1,1)
B. (1,1,*)
C. (*,1,1)
D. (0,*,*)
E. (1,1,1), (1,0,0), (0,1,0)

2. Boolean Algebra

How many ordered triples make the following expression TRUE?

$\overline{A\,\overline{AB}}\ \overline{BC} + \overline{C}$

A. 8
B. 7
C. 6
D. 5
E. 4

3. Data Structures

Find the internal path length of the binary search tree that is created from the following:

SPACESTATION

A. 48
B. 38
C. 36
D. 35
E. 30

4. Data Structures

Given an initially empty stack with the additional rule that, no letter is pushed onto the stack if it is identical to the top of the stack, what will the next popped item be?

PUSH(A), PUSH(T), PUSH(T), POP (X), PUSH(A), PUSH(C), PUSH(K), POP(X), POP(X), PUSH(O), PUSH(F), PUSH(T), POP(X), PUSH(H), PUSH(E), PUSH(E), PUSH(M), PUSH(P), POP(X), POP(X), PUSH(I), PUSH(R), POP(X), POP(X), PUSH(E), POP(X), POP(X)

A. A
B. F
C. H
D. O
E. nil

5. FSA's & Regular Expressions

Given the regular expression: **`[^dkp]*[aeiou][a-el-t]?(s|es|er)`**

Identify all of the following strings that are **not** accepted.

A.	dahlias	D.	aster	G.	clover
B.	roses	E.	azaleas	H.	lilacs
C.	lilies	F.	irises	I.	violets

A. A, E, G
B. D, E, G,
C. A, C, D, G
D. A, B, D, E, G
E. A, D, G